

Full Review: Beyond the Random Walk Sentiment Chaos and LLM Personas in Currency Forecasting AMRIT LAHARI

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Readability and Presentation

Section-by-Section Review: Writing and Presentation

Overall Narrative Assessment

Before diving into sections, the single biggest structural problem is that **the paper reads like three separate papers stitched together**. There is the base Phase A–D framework, then TEPC, then the Final-Phase stack. A reader following the narrative keeps asking: "Why does this next thing exist? What was wrong with the previous thing?" The transitions are weak and the motivations for each extension feel like post-hoc justifications rather than a coherent research story.

A strong paper has one thread: here is a problem, here is our insight, here is what we built, here is what we learned. This paper has three threads running in parallel, and they only loosely connect at the end.

Abstract

Problems

The abstract is the most important part of a paper and currently its weakest section. It is 280+ words of dense technical content with no narrative arc.

Issue 1: Acronym overload. In the first three sentences a non-expert encounters VMD, GDELT, XGBoost, GARCH, EGARCH, and TEPC with no context. The abstract should be readable by someone who has never heard of any of these.

Issue 2: Laundry list structure. The abstract lists components and numbers without explaining why those numbers matter. Saying "40% precision (double the random baseline)" is good, but then immediately jumping to "78.75% breakout accuracy on a strict 80-day raw-GDELT walk-forward" loses a reader who does not know what "breakout accuracy" or "raw-GDELT walk-forward" means.

Issue 3: Three separate stories. The abstract describes the base framework, then TEPC, then the Final-Phase stack, each in its own clause. A non-expert cannot tell which of these is the main contribution and which is secondary.

Issue 4: The deepest lesson is buried. The most surprising and generalizable finding — that a compressed memory snapshot beats frontier LLMs once persistence bias is removed — appears in the last sentence almost as an afterthought.

Suggested Rewrite Strategy

Structure the abstract in four sentences:

1. The problem and why it is hard.
2. What you propose at a high level (one sentence covering all three parts of the framework).
3. Your two or three sharpest empirical findings.
4. The central lesson.

Example draft:

Predicting USD/INR exchange rate movements is an open challenge: even sophisticated models rarely outperform a naïve random walk. We propose a multi-phase framework that combines news sentiment from 2.5 million GDELT articles, gradient-boosted classifiers, GARCH-family volatility modeling, chaos-inspired network features, and a debiased multi-persona LLM ensemble. Our danger-day volatility classifier doubles the random baseline precision (40% vs. 20%), political news sentiment leads rate movements by 9–12 days, and the full topology-chaos blend achieves 78% breakout accuracy under a strict two-day prediction embargo. Critically, once the trivial persistence baseline is removed, a tightly engineered memory snapshot outperforms all three frontier LLMs (Gemini 2.5 Flash, Claude Sonnet 4, GPT-5) on a 100-day backtest, suggesting that bias control matters more than model capacity for short-horizon forex forecasting.

This is shorter, flows naturally, and lands the central insight at the end.

Section 1: Introduction

Section 1.1 (Background and Motivation)

This is the strongest part of the introduction. The Meese-Rogoff framing is a good hook. The writing is clear.

Minor issue: The paragraph immediately invokes five citations in a row. This is academically correct but reads as citation-dropping rather than storytelling. Consider weaving citations into the narrative more naturally.

Section 1.2 (Problem Statement)

Issue: The formal mathematical definition of the three tasks appears on the second page of the paper, before the reader has any intuition for what is being attempted. The notation $x_t = [s_t, m_t, p_t]$ with $s_t \in \mathbb{R}^{80}$ will lose any non-specialist immediately.

Suggestion: Move the formal notation to Section 3 (Methodology). In the introduction, state the three tasks in plain English: "We ask three questions: which days will be highly volatile? Will the rate go up or down? And what will the exact value be?" The math can follow later when the reader is prepared for it.

Section 1.3 (Existing Work and Research Gaps)

Issue: This section lists five gaps but does not connect each gap to a specific contribution. A reviewer cannot easily map "Gap 3: No prior work has combined VMD with GDELT" to the part of the paper that addresses it.

Suggestion: Use a small table or inline parentheticals like "(addressed in Section 3.2)" after each gap. This creates a contract with the reader.

Section 1.4 (Proposed Approach)

Issue: This reads as a condensed version of the methodology section, duplicating content that will appear in detail later. It adds length without adding understanding.

Suggestion: Convert this to two or three sentences and a single clean diagram of the overall system. The reader does not need details here — just a map of the journey ahead.

Section 1.5 (Experimental Overview and Main Results)

Issue: Reporting specific numbers ("40% precision", " R^2 of 0.571") in the introduction is confusing. The reader has not yet seen the experimental setup, so these numbers have no frame of reference.

Suggestion: Remove specific numbers entirely. Write: "We find that classification substantially outperforms regression, that sentiment signals lead exchange rates by roughly ten days, and that simpler momentum models outperform deep learning on this dataset." Save the numbers for the results section.

Section 1.6 (Summary of Contributions)

Issue: Six bullet points describe contributions, but the bullets for TEPC and the Final-Phase are excessively long — each is a small paragraph. The contribution list should be scannable.

Issue: The ordering does not match the importance of contributions. The Final-Phase finding (bias control matters more than model capacity) is arguably the most important result for the field but appears last in a dense paragraph.

Suggestion: Trim each bullet to one sentence. Reorder to place the most important finding first. Add a sentence at the end explaining the relationship between the three main system components (base framework, TEPC, Final-Phase) so the reader understands the paper's architecture before diving in.

Section 2: Related Work

General

This section is competently written and covers the relevant literature. The "Gap addressed by this paper" subsections at the end of each part are a good structural choice.

Issue: The three subsections (news sentiment, exchange rate modeling, LLMs) are treated as independent silos. There is no connecting paragraph that explains how these three bodies of literature relate to each other and why this paper sits at their intersection.

Suggestion: Add a short paragraph at the end of Section 2 that synthesizes the three threads: "Prior work has separately explored sentiment signals for stocks, volatility models for forex, and LLMs for financial reasoning. Our framework is the first to combine all three in a unified pipeline for USD/INR specifically, and the first to apply debiased multi-persona prompting to the forex domain."

Section 3: Methodology

Section 3.1 (Overview)

Critical issue: The overview describes the Phase A–D pipeline but does not mention TEPC or the Final-Phase at all. A reader who has read the abstract knows these exist but then finds no mention of them in the overview. This is disorienting.

Suggestion: Expand Section 3.1 into a genuine system-level overview. Add a paragraph explaining: "We present three complementary frameworks. The base Phase A–D pipeline establishes the core approach. Section 3.9 extends this to a network-topology setting (TEPC). Section 3.10 introduces a stricter bias-controlled evaluation (Final-Phase). Each framework makes progressively stronger controls against evaluation bias."

This framing transforms the paper from three loosely connected systems into a coherent progression.

Figure 1 (Pipeline Diagram)

Issue: Figure 1 shows only the base Phase A–D pipeline. There is no figure showing how TEPC or Final-Phase relate to this. A reader has to hold three separate architectures in their head with no visual aid connecting them.

Suggestion: Add a high-level overview figure at the start of Section 3 showing all three frameworks side by side, with a brief label explaining what evaluation constraint each one tightens.

Section 3.2 (VMD)

This is clearly written. The intuition for why news affects only the noise component (IMF3) is well-motivated.

Minor issue: The paper states "We set $K = 3$, $\alpha = 2000$, and tolerance $\varepsilon = 10^{-7}$ " without justifying these choices. A reviewer will ask: why $K = 3$? Why not 4? Why $\alpha = 2000$?

Suggestion: One sentence of justification each: "We set $K = 3$ based on visual inspection of the spectral content, which clearly separates into trend, cycle, and noise bands. $\alpha = 2000$ was chosen to prevent mode mixing while preserving daily variation in IMF3."

Section 3.3 (Thematic Feature Engineering)

This section is well-written and the keyword lists are helpful.

Issue: The claim "3× higher correlation with exchange rate volatility compared to raw aggregation" appears in Figure 3's caption but is stated without statistical significance. A reviewer will ask whether this improvement is statistically meaningful or could be sampling noise.

Suggestion: Add a brief note on significance: "The improvement from 0.08 to 0.23 represents a 3× gain; a permutation test (N = 1,000) confirms this exceeds the 99th percentile of the null distribution."

Section 3.4 (Volatility Classification)

Issue: The threshold of 20% for "danger days" (top 20% of days by IIMF3I magnitude) is described as "heuristic" in the Limitations section but is not justified here.

Suggestion: Add a sentence: "The 20% threshold was selected to match empirically observed crisis-period frequency in the USD/INR series and to create a realistic 80/20 class imbalance; sensitivity to this threshold is discussed in the Limitations."

Sections 3.5 and 3.6 (EGARCH and Monte Carlo)

These sections are technically clear. No major issues.

Minor issue: Section 3.6 describes four Monte Carlo variants but Table 13 only shows results for some of them. The connection between methodology and results should be clearer.

Section 3.7 (Gemini V4)

Issue: Section 3.7 jumps straight to the architecture without motivating why LLMs are used at all. The theoretical justification (Section 3.8) comes after the architecture description, which is backwards: you should justify the approach before describing it.

Suggestion: Swap the order. Move Section 3.8 (Theoretical Grounding) to before Section 3.7, or at minimum add a paragraph at the start of 3.7 that says: "LLMs present a paradox for forecasting: they do not learn from live market data yet encode compressed representations of how analysts reason about markets. We argue in Section 3.8 that this makes them useful as structured information extractors, not predictors. With that framing, we describe our multi-persona architecture."

Issue: Table 1 (persona weights) contains fixed percentages (15%, 13%, etc.) but there is no explanation of how these weights were chosen. Were they tuned? Expert-assigned? Arbitrary?

Suggestion: Add a sentence explaining the provenance of these weights, or acknowledge that they are heuristically assigned and test sensitivity to them.

Section 3.8 (Theoretical Grounding)

This is actually one of the best-written sections. The three theoretical frameworks (compressed knowledge, Condorcet, bounded rationality) are clearly argued. The falsifiability paragraph is particularly strong.

Issue: The section title "Why LLM Personas Produce Meaningful Signals" reads defensively — as if the authors know a reviewer will push back. This framing puts the reader in an adversarial mindset.

Suggestion: Rename to something like "Theoretical Grounding for Multi-Persona LLM Forecasting" to make it read as a positive contribution rather than a defense.

Section 3.9 (TEPC)

Issue: TEPC is introduced with "As an extension we ask a different question..." This frames it as an afterthought. If TEPC is a genuine contribution (and the results suggest it is), it deserves a stronger motivation.

Suggestion: Open with the research gap: "The Phase A–D pipeline treats USD/INR as an isolated time series. However, exchange rates are embedded in a global network of correlated assets. We hypothesize that the topological structure of this network and its dynamical synchronization carry predictive information not available in any single node's time series. This motivates our TEPC framework."

Issue: The Lorenz oscillator equations (Equation 6) are introduced with minimal context for readers unfamiliar with chaos theory. The intuition for why a deterministic chaotic system is useful for financial prediction is not stated here (it appears later in the discussion).

Suggestion: Add two sentences before the equations: "Lorenz oscillators are deterministic dynamical systems that exhibit sensitivity to initial conditions (chaos). When coupled across the graph Laplacian, they produce synchronization dynamics: nodes that are well-connected tend to fall into phase. We use the degree of synchronization on any given day as a proxy for the market's collective coherence — a feature that the correlation matrix alone cannot capture."

Issue: The five-arm ablation is described in the Methodology section rather than the Experimental Setup. Ablation design belongs in Section 4.

Section 3.10 (Final-Phase)

Issue: Like TEPC, this section opens with "To stress-test..." which frames it as a secondary concern. The Final-Phase actually delivers the paper's most important methodological finding

(bias controls matter more than model capacity), so it should be framed as a primary contribution.

Suggestion: Open with a clear statement of the problem it solves: "A critical methodological concern runs through all prior work, including our own Phase A–D framework: using yesterday's closing price as a flat feature creates a persistence confound that silently inflates headline metrics. We address this with the Final-Phase architecture, which explicitly removes this crutch and enforces a strict two-day prediction embargo."

Issue: The description of the five experimental arms is very dense. A small table summarizing each arm (what features it uses, what it is designed to test) would make this much more scannable.

Section 3.11 (Design Choices and Justification)

Issue: This section lists seven design choices in a single dense paragraph. It is hard to parse.

Suggestion: Convert to a bullet list or a two-column table with "Choice" and "Justification" columns. This makes the design decisions immediately scannable for a reviewer.

Section 4: Experimental Setup

Section 4.1 (Datasets)

Generally clear. Table 2 is well-structured.

Issue: The paper uses "January 2025 – January 2026" as the data period. This is a single calendar year, and the Limitations section acknowledges this is short. A more prominent caveat here (not just in Limitations) would preempt reviewer concerns.

Section 4.2 (Models)

Issue: This is extremely long and reads as a configuration dump rather than a description. Parameters like `patience=10`, `dropout=0.2`, `max_depth=4` are important for reproducibility but not for understanding the experimental design.

Suggestion: Move all hyperparameter details to an appendix (call it "Appendix D: Hyperparameters and Implementation Details"). Keep Section 4.2 to a concise description of the model families and why each is included.

Section 4.3 (Evaluation Metrics)

Table 4 is excellent — one of the best tables in the paper. The description column is informative without being verbose.

Issue: The "Notes on the new metrics" paragraph is buried inside Section 4.3 rather than leading it. New readers need to understand the metrics before seeing the table.

Suggestion: Move the explanatory paragraph before the table, not after.

Section 4.4 (Baselines)

This section is well-organized. The nine numbered baseline categories clearly map to specific comparisons, and the closing paragraph explaining what each comparison isolates is very helpful.

Issue: The closing paragraph ("This baseline suite enables controlled comparisons...") is the most useful paragraph in the entire experimental setup section, but it is the last thing you read. Consider moving a condensed version of this insight earlier — perhaps as an opening paragraph to Section 4 — so readers understand the logic before they read the individual components.

Section 5: Results and Discussion

General

The results section has a structural problem that mirrors the methodology section: it presents three separate sets of results (Phase A–D, TEPC, Final-Phase) without a clear connective thread. A reader finishes Section 5.8 thinking the paper is over, then finds two more major sections of results.

Suggestion: Add a short introductory paragraph to Section 5 that maps out what the section covers: "We present results in three blocks. Sections 5.1–5.8 cover the base Phase A–D framework. Section 5.9 presents the TEPC ablation. Section 5.10 presents the Final-Phase strict-embargo ablation, which is the most carefully controlled evaluation in the paper. Section 5.11 synthesizes all findings."

Sections 5.1–5.2

Issue: Table 5 (feature correlations) appears in Section 5.1 but is referenced as context for Table 6 (danger-day results) in Section 5.2, with no table numbered as Table 5 appearing between them in the visible flow. The table numbering should be sequential and easy to follow.

Issue: The lead-lag finding (9–12 days) is one of the paper's most interesting results but is reported in a single paragraph in Section 5.3 without a figure. A cross-correlation plot showing this lag structure would make this finding far more compelling and visually memorable.

Suggestion: Add a figure showing the cross-correlation between political sentiment and exchange rate movements across lag values 0–20 days, clearly marking the peak at 9–12 days.

Section 5.5 (Regression Results)

Issue: The section title "Can traditional ML models predict exact exchange rate values?" is answered immediately with "no" (negative R^2). This is an important finding, but the writing slightly buries it by presenting the table without clear narrative commentary.

Suggestion: Open the section with a direct statement: "Exact price prediction is intractable on this dataset. Every regression model, regardless of complexity, produces negative R^2 on the test set, confirming that the noise component of exchange rates cannot be directly regressed. This validates our decision to focus on classification."

Section 5.7 (Ultimate Ensemble)

Issue: The stunning finding that the optimal ensemble converges to 95.5% MA Momentum + 4.5% GDELT Ridge is buried in the middle of the results. This deserves a callout box or bold emphasis. Deep learning researchers will be particularly surprised by this.

Suggestion: Add a "Key Finding" call-out or start the section with: "The most counterintuitive result in this paper: the optimized ensemble assigns near-zero weight to every deep learning model and nearly all complex approaches, converging to 95.5% moving average momentum plus 4.5% GDELT sentiment."

Section 5.9 (TEPC Ablation)

This section is well-structured. The three "findings" paragraphs are clear and easy to follow.

Issue: The observation that "the rank order is not stable across regimes" (TEPC-FULL is best in the 80-day window, CHAOS-ONLY is best in the 100-day window) is significant but is presented as a footnote-level observation. This is actually a meaningful limitation of TEPC — its best configuration is regime-dependent.

Suggestion: Flag this explicitly as a key finding, not just an observation: "This regime-sensitivity is a genuine limitation of the TEPC framework: the optimal configuration changes depending on the evaluation window and node universe, making deployment choices non-trivial."

Section 5.10 (Final-Phase)

Issue: The observation that all three frontier LLMs perform nearly identically (MAE prices 0.4433, 0.4439, 0.4508) is potentially the most interesting negative result in the paper, but it is underemphasized. It suggests that on this task, GPT-5, Gemini 2.5 Flash, and Claude Sonnet 4 are essentially interchangeable.

Suggestion: Give this finding its own short paragraph with explicit commentary: "The three frontier LLMs are statistically indistinguishable in our evaluation. This suggests that for structured short-horizon forecasting over a shared memory snapshot, LLM choice is not a critical design decision — prompt design and input engineering dominate model capacity."

Section 5.11 (Discussion)

This is the best-written section in the paper. The "Synthesis" and "Simpler models dominate" paragraphs are clear and appropriately frank.

Issue: The subsection "Strict embargo flattens the sentiment story" contains the paper's most important methodological lesson, but it is the second-to-last subsection. A reader who skims the paper will miss it.

Suggestion: Move the key lesson up. Consider starting Section 5.11 with the bias controls insight and working backwards to the other findings, rather than saving it for near the end.

Section 6: Conclusion

Issue: The conclusion largely repeats the abstract and Section 1.6. There is a very long list of contributions and future work but almost no reflection on what the authors now understand about the problem that they did not understand before.

Suggestion: Structure the conclusion differently:

1. One paragraph on what this paper set out to do.
2. One paragraph on what surprised you (MA Momentum wins, LLMs are interchangeable, bias controls dominate).
3. One paragraph on what this means for the field.
4. One or two sentences on future work (not a laundry list).

The current future work list has eight items. Trim to three most important and most specific.

Limitations Section

This is unusually thorough and honest, which is a strength. Most papers bury limitations. Keeping this section prominent reflects well on the authors.

Issue: The limitations are grouped under five bullet headings (dataset, methodological, model, computational, generalizability) but the individual bullet bodies are very dense. A reader scanning limitations cannot quickly identify the most important ones.

Suggestion: Add a single opening sentence identifying the most critical limitation: "The most significant limitation of this work is the short evaluation window (362 days), which constrains statistical power and precludes testing across different macro regimes."

Tables and Figures

General

Issue: Tables 14, 15, 16, 17, 18, and 19 appear in quick succession in Sections 5.9–5.10. This is dense. A reader is switching between six different result tables with different column structures, evaluation horizons, and metric definitions.

Suggestion: Consider consolidating Tables 14 and 15 (both TEPC ablations) into a single table with a "Window" column. Similarly, consolidate Tables 17 and 18 (Final-Phase 100-day and 30-day) into a single table. This reduces cognitive load.

Issue: Figures 6 and 7 (TEPC actual vs. predicted and rolling error) are referenced in the text but appear to be low-resolution screenshots rather than publication-quality figures. The axis labels are barely readable in the PDF.

Issue: There is no figure showing the Final-Phase results visually. Figure 8 shows a bar chart summary but it is small and the axes are compressed. A time-series plot of predicted vs. actual for the MEMORY-ONLY arm would significantly strengthen the paper's narrative.

Writing Style Issues (Global)

These apply throughout the paper:

Issue 1: Passive voice overuse. Many sentences use passive voice where active voice would be clearer. "EGARCH is fit on demeaned daily log returns" becomes "We fit EGARCH on demeaned daily log returns."

Issue 2: Inconsistent tense. The paper mixes present and past tense in the methodology section. Choose one (present tense is standard for methodology) and apply it consistently.

Issue 3: Jargon density. Terms like "walk-forward refit cadence", "abstaining range class", "strict-embargo walk-forward" are used without definition on first use. Each technical term should be defined the first time it appears.

Issue 4: Overly long sentences. Several sentences run to four or five clauses. For example the sentence beginning "We extend the framework along two complementary axes..." in the abstract runs for 60+ words. Break these into shorter sentences.

Issue 5: Acronym proliferation. The paper introduces TEPC, GDELT, VMD, IMF, EGARCH, GBM, RK4, MACRO-BASELINE, TOPOLOGY-ONLY, CHAOS-ONLY, TOPOLOGY-CHAOS, TEPC-FULL, STAT-ML-FULL, STAT-ML-NO-GDELT, STAT-ML-NO-MACRO, MEMORY-ONLY, RULE-PERSONAS. An acronym table or glossary in an appendix would help significantly.

Summary of Priority Edits

Ordered from highest to lowest impact:

Priority	Edit
1	Rewrite the abstract to have a narrative arc and end with the central lesson
2	Add a Section 3.1 overview that frames all three frameworks (base, TEPC, Final-Phase) as a coherent progression
3	Add a cross-correlation figure for the 9–12 day lead-lag finding
4	Move the Final-Phase motivation paragraph to the front of Section 3.10 and frame it as a primary contribution
5	Move hyperparameter details to an appendix; trim Section 4.2
6	Add a sentence to Section 5.7 calling out the MA Momentum finding explicitly
7	Add an acronym glossary appendix
8	Replace the Figure 2/3 screenshots with publication-quality vector figures
9	Trim the conclusion's future work list to three items
10	Add an opening paragraph to Section 5 mapping out the three result blocks

Consistency and Completeness

This is a deep scientific audit. I'll go through the paper methodically, organized by type of problem.

1. Data and Timeline Inconsistencies

1.1 Core Sample Size Contradiction

This is the most serious factual inconsistency in the paper. Three different numbers are given for the dataset size, and they cannot all be correct simultaneously.

- Abstract and Section 1.5: **"362 days of USD/INR exchange rate data"**
- Table 2 (Dataset Summary): **"256 records"** for USD/INR
- Table 22 (Project Statistics): **"256 daily"** exchange rate observations, but **"Training samples: 290 days, Test samples: 72 days"** — which sums to 362

If training + test = 290 + 72 = 362 but total observations = 256, the numbers are arithmetically impossible. 256 trading days in a calendar year is approximately correct (markets are open roughly 252 days), but 362 is close to calendar days. The paper never clarifies whether its sample counts are trading days or calendar days, and the inconsistency is never resolved. Every downstream claim that depends on sample size (statistical power, train/test split ratios) is affected.

1.2 The "2023" Label in Table 10

Table 10 is captioned **"Gemini V4 vs. V3 Performance Comparison (2023 Full Year, 262 Trading Days)"**. But the entire paper's data period is stated as **January 2025 – January 2026**. A year labelled "2023" in a 2025-data paper is either a copy-paste error from an earlier version or a genuine confusion about which data the Gemini comparison uses. This affects the credibility of the Gemini V4 results because a reader cannot determine whether the system was evaluated on the same data as all other models.

1.3 GPT-5 in the Evaluation

Tables 18 and 19 include results for **"openai/gpt-5-chat"** and the abstract lists GPT-5 as one of the frontier LLMs tested. The paper's stated evaluation period is 2025. GPT-5 was not publicly available at any point in 2025. The paper offers no explanation of what model is

actually being used here, when it was accessed, or whether "gpt-5-chat" refers to an actual GPT-5 release or a model alias. This is a significant reproducibility and validity concern.

2. Methodology Inconsistencies and Insufficiencies

2.1 VMD Look-Ahead Bias in Target Definition

Section 3.2 describes applying VMD to the full exchange rate series to produce IMF3. Section 3.4 then defines "danger days" as the top 20% of days by IMF3 magnitude. The problem: **VMD is computed on the entire dataset including the test period.** This means the test-set labels are defined using information that would not be available at prediction time. The IMF3 value for a test-period day is computed using future data points, since VMD is a global decomposition that optimizes across the full signal simultaneously. This is a form of look-ahead bias embedded in the target definition itself, and it is never discussed.

2.2 Independence Assumption in the Condorcet Argument

Section 3.8 argues that the multi-persona LLM system is grounded in the Condorcet Jury Theorem, which requires each agent to be **independent** and have accuracy $p > 0.5$. Two problems:

First, all nine personas use the same underlying Gemini model with different prompts. They share weights, training data, and systematic biases. Prompt-level differentiation does not produce statistical independence. The paper partially acknowledges this but then claims "inter-persona agreement rates range from 60–80%, confirming partial independence." Agreement rates of 60–80% actually indicate substantial herding rather than independence. If six to seven out of nine personas agree on every prediction, the effective sample size is much smaller than nine.

Second, the paper never demonstrates that any individual persona has accuracy $p > 0.5$ as required by the theorem. The theorem's guarantees do not apply if individual agents are at chance level.

2.3 Undefined Directional Thresholds

The paper defines three directional classes (UP/DOWN/STABLE) throughout but never states the thresholds that define STABLE. What size of daily movement counts as UP vs. STABLE? This definition directly affects the directional accuracy metric reported in every results table. Without this, the 41.2% three-class directional accuracy cannot be interpreted or compared to other work.

2.4 Ensemble Optimization Not Described

Section 5.7 states that the ultimate ensemble "converges to 95.5% MA Momentum + 4.5% GDELT Ridge." This is a key result, but the optimization methodology is entirely missing. What algorithm was used to find these weights? What objective function was optimized? What cross-validation scheme was applied? If the weights were optimized on the full training set and evaluated on the test set without held-out validation, there is data leakage.

2.5 Lorenz Initial Conditions Underspecified

Section 3.9 states that the initial conditions for the Lorenz oscillators are "deterministic functions of recent node volatility, drift, and range position" but never gives the functional form. Since Lorenz systems are chaotic, small changes in initial conditions lead to wildly different trajectories. The specific mapping from volatility/drift/range to initial conditions is critical for reproducibility and could significantly affect synchronization measurements. Without this, the TEPC results cannot be replicated.

2.6 Regime-Switching Monte Carlo Transition Probabilities Not Reported

Section 3.6 describes the recommended regime-switching Monte Carlo variant with two volatility states ($\sigma_H = 0.351\%$, $\sigma_L = 0.196\%$) and says "transition probabilities estimated from the news-volume threshold series." The transition probability matrix is never reported. This is essential for replication and for assessing whether the regime-switching behaviour is meaningfully different from a fixed-volatility GBM. The claim that this is the "recommended" variant is also unsupported — no comparative performance metric between the four Monte Carlo variants is shown.

2.7 Persistent-Laplacian Claim is Overstated

Section 3.9 calls the topology block "Persistent-Laplacian-style filtration" and the paper's full title includes "Topology-Enabled Predictions from Chaos." The paper itself acknowledges this is "a pragmatic graph-Laplacian-first approximation of a higher-order persistent Laplacian." True persistent Laplacians in the topological data analysis sense require simplicial complexes, boundary operators, and homology group tracking. What the paper actually implements is a multi-threshold spectral filtration of a weighted adjacency graph. Using "persistent-Laplacian" in the name and framing without this distinction prominently stated misleads readers familiar with TDA literature. The approximation may be practically effective, but it should not be called a persistent Laplacian.

2.8 Final-Phase Memory Snapshot Still Contains Price Information

Section 3.10 states the Final-Phase "explicitly removes the immediate previous close" and enforces a strict information embargo. However, the Volatility memory block within the shared snapshot includes "realized volatility, volatility ratio, trend pressure, reversal pressure."

Realized volatility is computed from recent price returns, and volatility ratio and trend pressure also implicitly encode recent price levels. The claim that the previous close is "never used as a flat feature" is technically correct but the spirit of the embargo — removing price persistence information — is partially violated through these derived features. This distinction should be explicitly acknowledged.

3. Results Inconsistencies

3.1 Gemini V4 Sample Errors Contradict the Stated Average

Table 11 shows four selected Gemini V4 January 2026 predictions with errors of 0.040%, 0.087%, 0.037%, and 0.051%. The average of these four is approximately 0.054%. Yet the abstract, Section 1.5, and Section 5.6 all claim "**~0.3% average prediction error.**" If the selected days are representative, the 0.3% figure is inconsistent. If they are cherry-picked best days, the table is misleading. The paper does not clarify this.

3.2 Rank Reversal in MEMORY-ONLY Between 30-day and 100-day Windows

The paper's headline claim is that MEMORY-ONLY is the best arm. In Table 17 (100-day backtest), MEMORY-ONLY has MAE price = 0.2421, beating STAT-ML-NO-GDELT at 0.2473. But in Table 18 (30-day LLM matrix), **STAT-ML-NO-GDELT (0.3188) outperforms MEMORY-ONLY (0.3295)**. This rank reversal is never commented on. The paper's conclusion that "MEMORY-ONLY dominates" is only true at the 100-day horizon. At 30 days, the result is reversed. This undermines the generality of the conclusion.

3.3 Three Frontier LLMs Have Identical Metrics

In Table 18, gemini-2.5-flash, claude-sonnet-4, and gpt-5-chat all have **exactly** the same directional accuracy (0.533) and sign F1 (0.6957). Identical metrics across three different models over 30 trading days is possible but unlikely without further explanation. Either the 30-day window is so small that all LLMs make the same directional calls, or there is an evaluation artefact. The paper calls this "nearly indistinguishable" but never investigates or comments on the exact equality.

3.4 TEPC Results Across Incomparable Settings

The abstract claims "the TEPC blend achieves 78.75% breakout accuracy on a strict 80-day raw-GDELT walk-forward, the chaos-only specification reaches 80% breakout accuracy with macro-F1 of 0.404 on a 100-day window." These are presented in the same sentence as if comparable, but they come from **different experimental settings**: different window lengths (80 vs. 100 days), different node universes (15 nodes including live GDELT vs. 12 nodes with FX crosses), and different data sources (raw-GDELT vs. live-pull). Quoting both in the same breath as evidence of TEPC's performance conflates two non-comparable experiments.

3.5 GARCH AIC Difference is Below Conventional Significance Threshold

Section 5.4 claims EGARCH achieves "the best fit (lowest AIC)" with AIC = 60.33, compared to GJR-GARCH at 61.36 and GARCH(1,1) at 60.45. The difference between EGARCH and GARCH(1,1) is only **0.12 AIC points**. Standard practice treats differences below 2 AIC points as negligible — both models are essentially equivalent fits. The claim of EGARCH superiority over GARCH(1,1) is not supported by this evidence. The paper should note that EGARCH and GARCH(1,1) are statistically indistinguishable here.

3.6 Feature Importance in Table 21 Does Not Sum to 100%

Table 21 (Top 10 Features in Hybrid EGARCH + XGBoost) lists importance values of 27.5, 15.8, 9.6, 8.2, 6.2, 4.4, 4.3, 4.1, 3.7, and 3.6. These sum to **87.4%**. The remaining 12.6% of feature importance is unaccounted for. This is not labelled as "top 10 only" and creates the false impression that these ten features fully explain the model.

4. Statistical Insufficiency

4.1 No Confidence Intervals or Significance Tests Anywhere

Every performance metric in the paper is reported as a point estimate with no error bars, confidence intervals, or statistical tests. This is a systemic problem. For example:

- "40% precision (double the random baseline)" — over 73 test days, what is the standard error of a 40% precision estimate? The two-sided 95% confidence interval for a proportion of 0.40 over 73 samples spans roughly ± 11 percentage points. The paper cannot claim 40% is meaningfully better than 20% without a significance test.
- "The debiasing techniques contributing a 2–3% improvement" — over the same test horizon, this is 2–3 days of different predictions. Not tested for significance.
- "9–12 day lead time between political sentiment and exchange rate movements" — no p-value, no confidence interval, no description of how this lag was estimated (cross-

correlation? Granger causality? Visual inspection?).

4.2 No Multiple Comparison Correction

Section 4.2 describes comparing 15+ models. With enough models being compared on the same test set, some will appear to outperform others by chance. No Bonferroni correction, Benjamini-Hochberg correction, or Diebold-Mariano test is applied anywhere. The claim that any specific model is "best" is therefore statistically ungrounded.

4.3 The EMH Violation Claim is Unsupported

Section 5.3 states the 9–12 day lead time "suggests a semi-strong form Efficient Market Hypothesis violation." This is a very strong claim. Demonstrating an EMH violation requires showing that a trading strategy based on this signal generates positive risk-adjusted returns after transaction costs, and that the result is statistically robust. The paper shows only a cross-correlation. The lead could reflect GDELT data collection lags (not all news enters GDELT instantaneously), holiday effects, or USD/INR-specific RBI intervention dynamics rather than a genuine pricing inefficiency. The word "suggests" softens the claim slightly, but it is placed prominently and remains misleading without proper testing.

4.4 Directional Accuracy Improvement Ranges Overlap

Table 10 shows Gemini V3 directional accuracy at 52–55% and Gemini V4 at 54–58%. The ranges overlap from 54% to 55%. The paper presents this as evidence that "debiasing improved directional performance," but overlapping ranges mean the two systems' performance distributions cannot be distinguished without a statistical test.

5. Internal Contradictions

5.1 TEPC Framing vs. TEPC Results

Section 3.9 opens with: "what if USD/INR is better modeled as one node in a coupled global macro–geopolitical network?" This framing implies TEPC is a superior model. But Table 14 shows **TEPC-FULL does not dominate on return MAE** — MACRO-BASELINE (0.002782) is essentially tied with TEPC-FULL (0.002746). More importantly, in the 100-day window (Table 15), CHAOS-ONLY actually outperforms TEPC-FULL on both breakout accuracy and macro F1. The paper's narrative positions TEPC-FULL as the headline result, but its own ablation shows that the topology block (which is the novel contribution) is not consistently additive.

5.2 "Bias Controls Matter More Than Model Capacity" vs. Model Recommendations

Section 5.11 concludes: "bias control may matter more than model capacity for short-horizon FX forecasting." Table 20 then recommends "Gemini Daily Predictor" for 1–3 day forecasting with expected error 0.04–0.25%. But the Gemini Daily Predictor is part of the Phase A–D system that the Final-Phase explicitly identifies as relying on the persistence confound. The recommended model table is based on results from the looser evaluation setup, directly contradicting the paper's own methodological conclusion. You cannot simultaneously conclude that the persistence baseline inflates Phase A–D metrics and recommend Phase A–D models as the best choice.

5.3 LLM Weight Convergence Argument vs. Final-Phase Findings

Section 3.8 argues: "if LLM personas produce pure noise, the adaptive weighting mechanism should converge to $w_{llm} \approx 0$ over time. We observe instead that w_{llm} stabilizes at 15–33%." This is presented as evidence that LLMs contribute non-redundant information.

But Section 5.10 finds that in the Final-Phase strict-embargo evaluation, frontier LLMs perform substantially worse than MEMORY-ONLY and only marginally better than rule-based personas. If the LLM adds non-redundant information as claimed in Section 3.8, why does the Final-Phase show it cannot beat a compressed memory snapshot? These two findings are in direct tension and the paper does not reconcile them.

5.4 Thematic Filtering Claim vs. Economy Theme Article Count

Section 3.3 argues that thematic filtering outperforms raw aggregation because themes capture domain-specific signals. But the Economy theme has only 139,953 articles vs. 594,924 for Policy and 560,938 for Conflict. The Economy theme, which most directly relates to exchange rate movements, is actually the smallest theme by article volume. If volume drives signal quality, the conflict and policy themes dominate by sheer scale. The claim that the four themes equally contribute to the 3x correlation improvement is inconsistent with this imbalance.

5.5 "No Look-Ahead Bias" Claim vs. Gemini V4 Warmup Period

Section 4.5 states: "Train/test splits are strictly chronological (80%/20%) to prevent look-ahead bias; no future data leaks into training at any stage." But Section 4.5 also notes: "The Gemini V4 system includes a 20-day warmup period." The warmup period implies the LLM system observes initial test-period data before making "live" predictions. If the warmup period

falls within the test set, this violates the stated no-look-ahead guarantee. The paper never clarifies whether the warmup days are drawn from training data or test data.

6. Incomplete Arguments

6.1 What Does the LLM Actually See?

Section 3.7.2 says "Nine specialized personas query Google Gemini AI independently" with "debiased prompts." The actual prompt content is never shown. For a paper making strong claims about prompt engineering and debiasing, this is a fundamental omission. Without seeing the prompts:

- The debiasing claims cannot be verified
- The "randomized direction labels" technique cannot be assessed
- The "symmetric case presentation" cannot be evaluated
- The paper is not reproducible

6.2 How the Statistical Model Weight Adapts is Incompletely Described

The formula $w_{\text{llm}} \in [0.15, 0.45]$ is said to "adapt based on entropy confidence, R^2 , volatility regime, and recent LLM value-added." But the functional form of this adaptation is never given. What formula maps these four inputs to a weight? This is central to understanding how the system behaves but is described only with a phrase.

6.3 The Lead-Lag Analysis Section Lacks Any Methodology

Section 5.3 is one paragraph: "Cross-correlation analysis reveals a critical finding: political sentiment leads exchange rate movements." It states a finding but gives no methodology. Which sentiment feature? What cross-correlation measure? What lags were tested? No figure. No table. No test statistic. No p-value. This is the paper's most widely cited potential discovery, and it is backed by the least methodological detail of any result in the paper.

6.4 Persona Accuracy Feedback Loop Not Validated

Debiasing technique 4 ("Persona accountability: Historical accuracy of each persona is fed back") implies a learning loop where personas improve over time. But the paper never shows whether individual persona accuracy improves across the evaluation period, whether the calibration data is in-sample or out-of-sample, or what happens in the first N days before any accuracy history is accumulated. The feedback loop is described as a design feature but its effectiveness is never demonstrated.

6.5 GDELT Data Quality is Unaddressed

The paper processes 2.5 million GDELT articles but never addresses GDELT's known quality issues: pre-computed tone scores are noisy, CAMEO event coding has high error rates in the 2+ coding positions, and GDELT duplicate-counts articles from the same event across many sources inflating NumMentions. These issues directly affect the thematic features. A paper using GDELT as its primary data source should at minimum acknowledge these limitations, which are well-documented in the political science literature.

Summary Table

Issue	Type	Severity
362 vs. 256 sample size contradiction	Factual	Critical
Table 10 labelled "2023" in a 2025 paper	Factual	High
GPT-5 used in 2025 evaluation	Factual	High
VMD look-ahead bias in target definition	Methodology	Critical
Condorcet independence assumption violated	Methodology	High
Directional threshold for STABLE undefined	Methodology	High
Ensemble optimization methodology missing	Methodology	High
Lorenz initial conditions underspecified	Reproducibility	High
Persistent-Laplacian term overstated	Framing	Medium
Realized volatility in embargo snapshot	Methodology	Medium
Gemini error 0.054% avg vs. 0.3% claim	Results	High
MEMORY-ONLY rank reversal at 30 days	Results	High
Three LLMs with identical metrics	Results	Medium
TEPC results across incomparable settings	Results	High
EGARCH vs. GARCH AIC difference insignificant	Results	Medium
Feature importance sums to 87.4%	Results	Low

Issue	Type	Severity
No confidence intervals anywhere	Statistics	Critical
No multiple comparison correction	Statistics	High
EMH violation claim unsupported	Claims	High
V3 vs. V4 accuracy ranges overlap	Statistics	Medium
TEPC narrative vs. topology block being weak	Contradiction	Medium
Table 20 recommendations vs. bias conclusion	Contradiction	High
LLM weight convergence vs. Final-Phase failure	Contradiction	High
Prompts not shown	Reproducibility	High
Lead-lag methodology entirely missing	Incompleteness	Critical

Novelty and Related Work

Now I'll synthesize all four searches into a comprehensive novelty review.

Novelty Review: "Beyond the Random Walk"

Overview of Novelty Claims in the Paper

The paper asserts the following novel contributions:

1. First debiased multi-persona LLM system for forex prediction
2. First combination of VMD with GDELT sentiment for exchange rate forecasting
3. First thematic filtering of GDELT for USD/INR specifically
4. Novel TEPC framework combining rolling graph-Laplacian filtration with coupled Lorenz oscillators
5. Novel Final-Phase bias-controlled memory architecture with strict embargo

Each is assessed below against the literature found.

Claim 1: "First Debiased Multi-Persona LLM System for Forex Prediction"

Verdict: Substantially overstated. Not adequately cited.

This is the paper's headline novelty claim, and it is the one with the most directly relevant prior art that the paper fails to cite.

Uncited prior work on multi-persona / multi-agent LLM financial forecasting

Leveraging LLMs for Institutional Portfolio Management: Persona-Based Ensembles (arXiv 2411.19515, November 2024, University of Tokyo). This paper explicitly uses distinct LLM personas (economist, risk manager, etc.) in an ensemble for predicting price movements using macroeconomic indicators, with aggregation via mode voting. This is the closest single prior work to the paper's multi-persona Gemini system. It is not cited.

TradingAgents: Multi-Agents LLM Financial Trading Framework (arXiv 2412.20138, December 2024). Simulates a full trading firm with seven specialized LLM agents including a Fundamentals Analyst, Sentiment Analyst, Technical Analyst, and Risk Manager, with a Bull/Bear researcher debate mechanism. The structural similarity to the paper's nine-persona Gemini system is substantial. Not cited.

FinCon: A Synthesized LLM Multi-Agent System (NeurIPS 2024, arXiv 2407.06567). A manager-analyst hierarchy with seven specialized analyst agents and a conceptual verbal reinforcement mechanism for sequential financial decision-making, accepted at a top-tier venue. Not cited.

Wisdom of the Silicon Crowd (arXiv 2402.19379, February 2024, Philip Tetlock group). Tests a crowd of twelve LLMs as diverse "personas" on binary forecasting questions against 925 human forecasters. Directly establishes the "wisdom of crowds via LLM diversity" paradigm that the paper's Section 3.8 Condorcet argument relies on. The paper cites Condorcet (1785) but not this directly relevant empirical validation. Not cited.

Prompting for Policy: Macroeconomic Forecasting with Synthetic LLM Personas (arXiv 2511.02458, November 2025). Evaluates 2,368 synthetic LLM personas for macroeconomic forecasting across 50 quarterly rounds of the ECB Survey of Professional Forecasters. Large-scale empirical validation of persona-based LLM ensembles for economic forecasting. Not cited.

PolySwarm (arXiv 2604.03888, April 2026). Deploys fifty diverse LLM personas with confidence-weighted Bayesian aggregation specifically for prediction market trading. Directly parallel architecture. Not cited.

What the paper does cite

The paper cites Lopez-Lira and Tang (2023) for GPT-based financial sentiment and Li et al. (2024) for "multi-agent LLM systems for financial decision-making" (with a placeholder arXiv ID: arXiv:2402.xxxxx). This citation is suspicious — a paper cited as "Li et al. (2024)" with a placeholder arXiv ID "2402.xxxxx" is either unverifiable or a fabricated reference. It cannot be located with the information provided.

Assessment

The individual components of the paper's LLM contribution (multi-persona design, ensemble aggregation, debiasing) all have direct prior art published before the paper's stated data period. The paper's claim of being "first" cannot stand without either demonstrating that those prior works were not available at submission time, or reframing the novelty as the specific combination applied to forex rather than the paradigm itself.

Claim 2: "First to Combine VMD with GDELT Sentiment for Forex"

Verdict: The narrow combination may be new, but both halves are well-precedented and the paper misrepresents the prior art.

VMD for forex is not new

Forecasting exchange rate using VMD and entropy theory (Physica A, 2018). VMD applied to exchange rate decomposition for forecasting. Six years prior to this paper.

A Novel VMD-Based LSTM for Foreign Exchange Prediction (Springer LNCS, 2022/2023). Combines VMD with LSTM for forex prediction, directly analogous to the paper's Phase A–C structure.

A Novel Hybrid Deep Learning Method for Exchange Rate Prediction using VMD (MDPI Risks, 2024). VMD augmented deep learning for monthly FX forecasting.

VMD + GARCH + LSTM hybrid for stock price forecasting (Wiley IJMMS, 2025). Very close architectural precedent.

The paper cites Dragomiretskiy and Zosso (2014) for the original VMD algorithm but does not cite any of these VMD-for-forex applications. A reader has no way to assess what is actually new in the VMD application.

Sentiment + VMD combination is not new either

An enhanced risk-aware multi-task learning framework with dual-source sentiment and multi-scale VMD decomposition (Expert Systems with Applications, Elsevier, 2025).

Combines dual-source news and forum sentiment with VMD decomposition for financial time series prediction. This is precisely the sentiment-plus-VMD combination the paper claims as novel. Not cited.

Fluctuation correlation between investor sentiment and stock index using VMD-LSTM (Economic Analysis and Policy, ScienceDirect, 2023). VMD decomposition combined with investor sentiment indices and LSTM for stock prediction. Directly relevant. Not cited.

What is genuinely narrow-novel

No paper was found that specifically uses GDELT (as opposed to general sentiment sources) together with VMD for any financial prediction task. This specific intersection may be genuine, but the paper must frame it as "GDELT-specific" rather than "first to combine VMD with sentiment for forex." The current framing is misleading.

Claim 3: "First to Use GDELT with Thematic Filtering for USD/INR"

Verdict: Both the GDELT application and the USD/INR sentiment work have prior art. The intersection may be narrow-novel but is not presented accurately.

GDELT for financial prediction is not new

Macroeconomic Forecasting through News, Emotions and Narrative (Tilly, Ebner, Livan; Expert Systems with Applications, Elsevier, 2021). Uses GDELT news data with thematic filtering methodology and BiLSTM for macroeconomic forecasting. This paper explicitly introduces thematic data filtering on GDELT articles — the same paradigm the paper claims as its own. Published four years earlier in a high-impact journal. Not cited.

Interpretable Machine Learning for Macro Alpha: A News Sentiment Case Study (arXiv 2505.16136, May 2025). Ingests daily GDELT news, applies FinBERT for sentiment scoring, and aggregates Goldstein scores and tone features into XGBoost classifiers for EUR/USD

and USD/JPY prediction. Achieves Sharpe ratios of 5.87 and 4.65 respectively. This is a direct GDELT-for-forex prediction paper. Not cited.

Using the GDELT Dataset to Analyse EU Sovereign Bond Markets (Consoli et al., Springer LNCS, 2021). GDELT sentiment features (including Goldstein scores and article volume, the same features this paper uses) applied to financial market prediction. Not cited.

Measuring News Media Sentiment Using GDELT for Chinese Stock Markets (Pacific-Basin Finance Journal, Elsevier, 2022). Constructs multiple GDELT-based sentiment time series (Tone, Optimism, Attention) for financial prediction. Not cited.

News sentiment for USD/INR is not new

Impact of News Sentiment on Foreign Exchange Rate Prediction (Tadphale et al., IEEE CONIT, 2023). Uses FinBERT for news sentiment classification specifically for USD/INR prediction and demonstrates 15% error reduction. Direct prior work on the exact currency pair. Not cited.

Multiple additional papers on USD/INR forecasting with ML (LSTM, Random Forest, ANN-GA hybrids) are also not cited, meaning the USD/INR baseline literature review is incomplete.

Claim 4: TEPC — Novel Combination of Persistent-Laplacian Features and Coupled Lorenz Oscillators

Verdict: Mixed. The coupled Lorenz engineering is moderately novel; the Laplacian component has clear prior art that is not cited.

Persistent Laplacian features for financial markets is not new

Laplacian Spectra of Persistent Structures in Taiwan, Singapore, and US Stock Markets (Entropy, MDPI, 2023). Applies persistent Laplacian spectra, including Fiedler vectors and spectral gaps, to stock market crash detection across major indices. This is direct prior art for using persistent Laplacian spectral features in financial forecasting. The paper claims TEPC's persistent-Laplacian features are novel, but this 2023 paper directly contradicts that. Not cited.

The Persistent Laplacian for Data Science: Evaluating Higher-Order Spectral Representations (ICML 2023, proceedings.mlr.press). The foundational machine learning paper establishing persistent Laplacian as a computational tool. The paper cites no TDA or persistent Laplacian methodology paper at all, despite building a contribution around it.

Graph Signal Processing for Global Stock Market Volatility Forecasting (arXiv 2410.22706, 2024). Magnetic Laplacian combined with a HAR model for volatility forecasting across twenty-four global stock indices using spillover networks. Directly relevant to the TEPC graph-based approach. Not cited.

Algorithms for Learning Graphs in Financial Markets (arXiv 2012.15410, 2020). Learns Laplacian-constrained graphs specifically from foreign exchange networks. Not cited.

Lorenz oscillators for financial forecasting

FOREX Rate Prediction Using Chaos, Neural Network and PSO (Springer, ICSI 2014). Combines chaos theory features with neural networks and PSO for forex rate prediction. Prior art for chaos-derived features in forex. Not cited.

Expectations, Price Fluctuations and Lorenz Attractor (Olkhov, MPRA, 2018). Shows that economic buy-sell transaction dynamics reduce to Lorenz attractor form. Theoretically establishes that Lorenz dynamics are intrinsic to price fluctuations. Not cited.

Revisiting Paul de Grauwe's Chaotic Exchange Rate Model (Open Economies Review, Springer, 2022). Chaotic exchange rate dynamics with endogenous chaos. Not cited.

What is moderately novel in TEPC

Engineering coupled Lorenz oscillators across a time-varying graph Laplacian and extracting their synchronization state as an ML feature vector for forex forecasting does not appear in any found paper. This engineering application is the most plausibly novel component of TEPC, though it is built on well-precedented theoretical foundations that are not properly cited.

Claim 5: Final-Phase Memory-First Architecture

Verdict: The specific architecture may be novel but the bias-control motivation is established prior art.

Evaluating LLMs in Finance Requires Explicit Bias Consideration (arXiv 2602.14233, February 2026). Identifies five types of bias in financial LLM papers including look-ahead bias from using the previous close. The paper's Final-Phase design addresses exactly these identified biases, but does not cite this work.

Look-Ahead Bias in Large Language Models: Implications and Applications in Finance (Alonso, SSRN, 2024). Specifically quantifies look-ahead bias in LLM-based financial backtesting — directly relevant to the Final-Phase's motivation. Not cited.

Anchoring Bias in Large Language Models: An Experimental Study (Journal of Computational Social Science, 2025). Demonstrates anchoring bias in GPT-4, Gemini Pro, and Claude 2 and shows CoT prompting mitigates it. More rigorous empirical support for the debiasing techniques the paper uses than its current self-citations. Not cited.

Systematic Citation Gaps

The paper's reference section reveals structural omissions that go beyond individual papers.

Missing venue coverage

The paper cites no papers from NeurIPS, ICML, ICLR, or ACL/EMNLP from 2023–2025. Given that LLM-based financial forecasting has exploded as a research area at exactly these venues during this period (FinCon at NeurIPS 2024, TradingAgents widely disseminated, ICML 2023 persistent Laplacian paper), the absence of any top-ML-venue citation is a significant gap.

The "Li et al. (2024)" phantom citation

The paper cites "Li et al. (2024). Large language models for financial decision-making: A multi-agent perspective. arXiv preprint arXiv:2402.xxxxx" with a literal "xxxxx" placeholder as the arXiv identifier. This is either an unfinalised draft reference that was never properly filled in, or a fabricated citation. Neither is acceptable.

Missing survey papers

The paper is missing citations to all major recent surveys of LLMs in finance:

- "The New Quant: A Survey of LLMs in Financial Prediction and Trading" (arXiv 2510.05533, 2025)
- "Large Language Model Agent in Financial Trading: A Survey" (arXiv 2408.06361, 2024)
- "Large Language Models in Finance: A Survey" (arXiv 2311.10723, 2023)

Given that the paper claims several "firsts," the absence of any engagement with these survey papers — which systematically catalogue what exists — is a significant weakness.

What Is Genuinely Novel

After accounting for all prior art found, the following components have a reasonable novelty claim:

Component	Novelty Assessment
GDELT + VMD + USD/INR as a joint system	Narrow novelty — the specific triple combination has not been found in one paper
Nine-persona debiased Gemini system with six specific debiasing techniques applied jointly to a forex task	Moderate novelty — multi-persona LLM for finance exists, but the specific debiasing operationalization in forex is not found as a prior single system
Coupled Lorenz oscillators across a time-varying graph Laplacian as ML features	Moderate novelty as an engineering contribution — not found as a prior published pipeline
Final-Phase embargo design with shared compressed memory snapshot	Moderate novelty — the specific architecture is new even if the bias-control motivation is not
Empirical finding of 9–12 day political sentiment lead for USD/INR	Potentially novel empirical finding for this specific currency pair, though methodologically underspecified

Summary Assessment

The paper makes novelty claims that are significantly stronger than what the evidence supports. The core problem is not that the paper lacks genuine contributions — it does have real technical substance — but that the claims are framed as absolute firsts when each component has meaningful prior art, and the paper has failed to engage with a substantial body of directly relevant recent work. A reviewer familiar with the 2023–2025 LLM-in-finance literature would flag the missing TradingAgents, FinCon, arXiv 2411.19515, the GDELT-for-forex papers, the VMD-for-forex papers, and the persistent Laplacian papers as necessary related work. The placeholder citation "arXiv:2402.xxxxx" is an additional red flag about the thoroughness of the literature review.

The appropriate framing for the genuine contributions is not "first ever" but rather "first applied to USD/INR" and "first to jointly combine these specific components." That is a narrower but defensible novelty claim.

Conference Review — EMNLP 2026

EMNLP 2026 Review: "Beyond the Random Walk: Sentiment, Chaos, and LLM Personas in Currency Forecasting"

Part 1: Conference-Style Review

1. Paper Summary

This paper presents a multi-phase framework for USD/INR exchange rate forecasting that integrates three complementary systems. The base framework (Phases A–D) applies Variational Mode Decomposition (VMD) to isolate the news-driven volatility component of USD/INR, engineers thematic features from 2.5 million GDELT news articles across four market-relevant themes (Economy, Conflict, Policy, Corporate), trains XGBoost classifiers for volatility and directional prediction, and introduces a debiased nine-persona Gemini AI ensemble with six prompt-level debiasing techniques. Two extensions are layered on top: a Topology-Enabled Predictions from Chaos (TEPC) module that treats USD/INR as a node in a global macro network and combines rolling graph-Laplacian filtration with coupled Lorenz oscillators as directional features; and a Final-Phase memory-first architecture that enforces a strict two-day prediction embargo and feeds a compressed market memory snapshot to statistical, rule-based, and frontier-LLM (Gemini 2.5 Flash, Claude Sonnet 4, GPT-5) models under identical input conditions. Experiments on 2025 data report 40% danger-day precision (vs. 20% random baseline), 54–58% directional accuracy for the Gemini V4 system, 78.75% breakout accuracy for the TEPC persona blend, and 61% directional accuracy for the MEMORY-ONLY Final-Phase arm. The paper's most important finding is that a tightly engineered compressed memory snapshot outperforms all three frontier LLMs and full feature stacks on price MAE under the strict embargo, suggesting bias control dominates model capacity for short-horizon forex forecasting.

2. Strengths

- **Practically motivated and timely problem.** USD/INR forecasting using large-scale news data and LLM reasoning is a well-motivated, underexplored intersection. The managed-float dynamics of the rupee make it a genuinely interesting case study distinct from heavily studied pairs like EUR/USD.
 - **Layered experimental rigor.** The paper's most disciplined contribution is the Final-Phase design: removing the previous close, enforcing a two-day embargo, and feeding an identical compressed memory snapshot to every model family is methodologically sound and addresses a real confound (persistence inflation) that most comparable papers ignore.
 - **Debiasing framework is practically useful.** The six-technique debiasing pipeline for LLM prompting (symmetric case presentation, randomized order, explicit uncertainty, persona accountability, bias feedback, pips constraint) is a concrete, operationalized contribution to prompt engineering for financial NLP that would be of independent interest to the community.
 - **Honest ablation design.** The five-arm ablation in both TEPC and Final-Phase is well-structured, allowing the contribution of each component to be isolated. The paper does not cherry-pick the best arm but reports the full table, including cases where the proposed method does not win.
 - **Empirically transparent about model failures.** The paper forthrightly reports that deep learning baselines (LSTM, GRU, Attention) produce negative R^2 and that the optimal ensemble degenerates to 95.5% moving average momentum. This honesty is commendable and adds scientific value.
 - **Broad baseline coverage.** The comparison spans naïve baselines, classical time series, GARCH-family volatility models, ML approaches, deep learning, and multiple LLM backends — among the most comprehensive baseline suites seen in financial NLP work.
 - **The Final-Phase frontier-LLM result is interesting.** The finding that Gemini 2.5 Flash, Claude Sonnet 4, and GPT-5 are effectively indistinguishable on this task, and all inferior to a compressed memory snapshot, is a non-obvious, publishable negative result that speaks directly to the hype around LLMs in financial forecasting.
-

3. Weaknesses

Critical issues (would prevent acceptance in current form):

- **Irreconcilable sample size contradiction.** The abstract claims 362 days of USD/INR data, Table 2 reports 256 exchange rate records, and Table 22 lists 290 training + 72 test = 362 samples. These cannot simultaneously be true. Every downstream statistical claim depends on the sample size, making it impossible to assess whether any result is meaningful.
- **Table 10 caption reads "2023 Full Year, 262 Trading Days"** in a paper whose stated data period is January 2025 – January 2026. It is unclear whether the Gemini V4 comparison was conducted on 2023 or 2025 data, directly undermining the primary LLM result.
- **Phantom citation.** Li et al. (2024) is cited as "arXiv:2402.xxxxx" — a literal placeholder identifier that cannot be verified or located. This is unacceptable for published work.
- **VMD look-ahead bias.** VMD is applied globally to the full time series before train/test splitting, making the IMF3 noise component — which serves as the prediction target and the basis for danger-day labels — a function of future test-period data. This is a form of target leakage embedded in the experimental design.
- **Lead-lag claim entirely unsubstantiated.** The 9–12 day political sentiment lead, prominently advertised as a "potential semi-strong EMH violation," is supported by a single paragraph with no methodology, no figure, no test statistic, and no p-value. It cannot be evaluated as presented.
- **No confidence intervals or significance tests anywhere.** All metrics are point estimates. Over 72–100 test days, a 3-percentage-point directional accuracy difference corresponds to 2–3 days and is almost certainly not statistically significant. No Diebold-Mariano test, no bootstrap confidence intervals, no permutation tests.
- **GPT-5 is named as a tested model.** GPT-5 was not publicly available during the 2025 evaluation period. The paper provides no explanation of what model is actually used under this identifier, creating a reproducibility and validity concern.

Significant issues:

- **Critical related work missing.** The paper misses directly relevant prior work that challenges its novelty claims: TradingAgents (arXiv 2412.20138, multi-agent LLM trading with specialized roles), FinCon (NeurIPS 2024, seven specialized LLM analyst agents), arXiv 2411.19515 (persona-based LLM ensembles for portfolio management), Tilly et al. (2021, GDELT with thematic filtering for macroeconomic forecasting), arXiv 2505.16136 (GDELT with Goldstein scores for EUR/USD prediction), VMD-for-forex papers (2018–2024), and the persistent Laplacian for stock markets paper (Entropy 2023). The claim of being "first" is not defensible given these omissions.

- **Condorcet independence assumption violated without acknowledgment.** All nine personas use the same underlying Gemini model. Agreement rates of 60–80% indicate substantial herding rather than the near-independence required for Condorcet guarantees. The theoretical grounding in Section 3.8 overstates the case.
- **TEPC "persistent Laplacian" framing is misleading.** The paper implements a multi-threshold spectral filtration of a weighted adjacency graph, which is a significant departure from true persistent Laplacian methods (requiring simplicial complex construction and boundary operators). Using "persistent Laplacian" in the paper title's spirit and in the TEPC name misleads readers familiar with TDA.
- **MEMORY-ONLY rank reversal is unacknowledged.** The headline claim that MEMORY-ONLY is the best arm holds at 100 days but reverses at 30 days, where STAT-ML-NO-GDELT outperforms it. This undermines the generality of the conclusion.
- **Table 20 (model recommendations) directly contradicts the paper's main methodological lesson.** The paper concludes that bias controls matter more than model capacity, yet recommends the persistence-confounded Gemini Daily Predictor for 1–3 day forecasting based on inflated metrics.
- **Three frontier LLMs achieve exactly identical directional accuracy and Sign F1 in Table 18.** This is not commented on and raises questions about the evaluation mechanism over 30 days.
- **No prompts are shown.** For a paper whose NLP contribution centers on prompt debiasing, the complete absence of any prompt text makes the core contribution unverifiable and non-reproducible.

Minor issues:

- The paper reads structurally as three loosely connected systems without a coherent motivating arc. The relationship between Phase A–D, TEPC, and Final-Phase is never clearly framed in the introduction or methodology overview.
- EGARCH AIC improvement over GARCH(1,1) is 0.12 points — below the conventional significance threshold of 2.0.
- Feature importance table (Table 21) sums to 87.4%, not 100%.
- No directional threshold for the STABLE class is ever defined.
- Figure quality is insufficient for publication — axes are unreadable.

4. Overall Recommendation

Weak Reject (Score: 2/5, Confidence: 4/5)

Justification. This paper addresses a legitimate and timely problem and contains a genuine empirical contribution — particularly the Final-Phase architecture with its strict embargo design and the negative finding that frontier LLMs are inferior to a well-engineered memory snapshot. These results are worth publishing.

However, the paper cannot be accepted in its current form for three reasons. First, the data inconsistency (362 vs. 256 observations) and the "2023" table caption create factual ambiguity that undermines trust in all reported results. Second, the phantom citation and absent prompt text make the primary NLP contribution unverifiable. Third, the paper systematically misses 2023–2025 related work at NeurIPS, arXiv, and domain journals that directly challenges its novelty claims, and the "first ever" framing is not defensible. With substantial revision — particularly resolving the data inconsistency, adding the missing related work, showing the prompts, adding significance tests, and fixing the internal contradictions — this paper could reach the acceptance threshold. The final-phase results and the debiasing methodology in particular have genuine conference-level value.

Part 2: Comprehensive Revision Plan

Tier 1: Must-Fix Before Resubmission (Fatal Issues)

These issues would cause rejection at any venue regardless of other strengths.

R1. Resolve the Sample Size Contradiction

Problem. 362 (abstract) vs. 256 (Table 2) vs. $290+72=362$ (Table 22) — these are incompatible.

Required action.

- Audit the actual number of trading days in the dataset.
- Decide whether the paper reports calendar days or trading days and apply this consistently everywhere.
- Update all mentions of sample size (abstract, Section 1.5, Table 2, Table 22, Appendix C) to

a single consistent number.

- Recheck all train/test split ratios, which are stated as 80%/20% — verify these hold under the corrected count.

R2. Fix or Remove the "2023" Label in Table 10

Problem. Table 10 is captioned "2023 Full Year, 262 Trading Days" in a paper whose data period is 2025.

Required action.

- Determine which year the Gemini V3 vs. V4 comparison was actually run on.
- If it was run on 2025 data: correct the caption.
- If it was run on 2023 data as part of an earlier pilot: clearly separate this as a historical pilot result, note it uses different data from all other experiments, and explain why.
- Do not present results from different years in the same results section without explicit separation.

R3. Fix or Remove the Phantom Citation

Problem. "Li et al. (2024). arXiv:2402.xxxxx" is an unresolvable citation.

Required action.

- Identify the actual paper this was intended to cite and provide the correct arXiv ID or DOI.
- If no such paper exists, remove the citation and replace it with the appropriate alternative from the missing related work identified below.
- Conduct a full citation audit: verify every arXiv ID in the reference list is a real, accessible paper.

R4. Clarify GPT-5 Availability

Problem. GPT-5 is listed as a tested model in a paper evaluated on 2025 data.

Required action.

- Provide the exact OpenRouter model string used (e.g., `openai/gpt-5-chat`), its release date, and the dates on which the paper ran queries against it.
- If this was accessed after general availability in late 2025 or 2026, state this explicitly.
- If "gpt-5-chat" refers to a different model, identify it correctly.

R5. Fix the VMD Look-Ahead Bias

Problem. VMD is applied globally to the full time series, making test-set labels functions of future data.

Required action.

- Rerun VMD in a walk-forward manner: fit VMD only on the training window, then apply the learned mode parameters to decompose new observations as they arrive.
 - Alternatively, fit VMD on training data only and report the decomposition quality on held-out data separately.
 - Recompute all danger-day labels and IMF3-based features under the corrected walk-forward scheme.
 - Report how much (if at all) performance changes under the corrected protocol. If results degrade, report this honestly — it is an important methodological finding in itself.
-

Tier 2: Required for Scientific Soundness

These issues must be addressed to make the results interpretable and the claims defensible.

R6. Add Statistical Significance Tests Throughout

Problem. All metrics are point estimates over 72–100 test days with no standard errors.

Required action.

- For classification metrics (precision, recall, F1, directional accuracy): report Wilson score 95% confidence intervals for proportions.
 - For regression metrics (MAE, RMSE): report bootstrap 95% CIs (2000 bootstrap replicates, stratified by month to preserve temporal ordering).
 - For model comparisons: apply the Diebold-Mariano test for equal predictive accuracy between the proposed model and each baseline.
 - For the TEPC vs. MACRO-BASELINE comparison and the MEMORY-ONLY vs. LLM comparison: apply McNemar's test for directional accuracy differences.
 - Apply Bonferroni or Benjamini-Hochberg correction for the multi-model comparison in Table 12 (15+ models).
-

R7. Document and Show the LLM Prompts

Problem. The core NLP contribution (debiased multi-persona prompting) is entirely unverifiable without the actual prompts.

Required action.

- Add an appendix ("Appendix E: LLM Prompt Templates") showing the complete prompt for at least one persona (e.g., the Macro Analyst), with placeholders marking where dynamic data (statistical model output, confidence intervals, bias history) are inserted.
 - Show how the bull/bear randomization is implemented in the prompt template.
 - Show how the pips constraint is enforced in the system prompt.
 - For the Final-Phase LLM personas: show the memory snapshot JSON schema and the output JSON schema (direction, magnitude, confidence, thesis, risk_flags).
-

R8. Provide Methodology for the Lead-Lag Finding

Problem. The 9–12 day political sentiment lead is stated without any supporting methodology, figure, or test.

Required action.

- Add a subsection in Section 5.3 describing the cross-correlation methodology: which sentiment feature (TonePolicy? ToneAll?), which exchange rate series (IMF3? raw returns?), what lag window tested, what significance threshold used.
 - Add a cross-correlation figure plotting the correlation coefficient against lag values from 0 to 20 days, with 95% confidence bands under the null hypothesis of no correlation (computed by permutation).
 - Apply a Granger causality test (VARs at lags 5, 10, 15 days) to distinguish predictive leading from contemporaneous correlation.
 - Soften the "semi-strong EMH violation" claim to "is consistent with a semi-strong EMH violation, pending causal analysis" if Granger tests are not available.
-

R9. Acknowledge and Address the MEMORY-ONLY Rank Reversal

Problem. MEMORY-ONLY is the headline best arm at 100 days but is outperformed by STAT-ML-NO-GDELT at 30 days, unremarked.

Required action.

- Add a paragraph in Section 5.10 explicitly discussing this rank reversal.
- Investigate whether the reversal is a function of window length or the different evaluation dates (different macroeconomic regimes).
- Qualify the main conclusion: "MEMORY-ONLY is the best arm at the 100-day horizon; at

shorter windows the macro-only statistical model shows competitive or superior MAE, suggesting the memory architecture's advantage grows with evaluation length."

R10. Fix the Table 20 Recommendation Contradiction

Problem. Table 20 recommends the Gemini Daily Predictor (Phase A–D, persistence-confounded) while the paper's main lesson is that persistence confounds inflate Phase A–D metrics.

Required action.

- Either remove Table 20 entirely, or
- Add a footnote explicitly noting: "These expected errors reflect the looser evaluation setup where the previous close is available. Under the strict embargo (Final-Phase), the recommended system for all horizons below one month is the MEMORY-ONLY architecture reported in Table 17."
- Update the recommendation for 1–3 days to MEMORY-ONLY with the strict-embargo MAE figure.

Tier 3: Novelty and Related Work

R11. Revise All "First" Claims and Add Missing Related Work

Problem. Multiple directly relevant prior works are missing. Novelty is overstated.

Required action — new citations to add:

Paper	Why it must be cited
TradingAgents (arXiv 2412.20138, 2024)	Multi-agent LLM trading with specialized personas; directly challenges "first" claim
FinCon (NeurIPS 2024)	Seven specialized LLM analyst agents for finance; top-tier venue
arXiv 2411.19515 (2024)	Persona-based LLM ensembles for portfolio management; same paradigm
Tilly et al. (Expert Systems with Applications, 2021)	GDELT with thematic filtering for macroeconomic forecasting; challenges Phase

Paper	Why it must be cited
	B novelty
arXiv 2505.16136 (2025)	GDELT + Goldstein scores for EUR/USD prediction; direct competitor
Wisdom of the Silicon Crowd (arXiv 2402.19379, 2024)	Crowd of LLM personas for forecasting; grounds Section 3.8 Condorcet argument
VMD-for-forex papers (Physica A 2018, Springer LNCS 2023, MDPI 2024)	Prior VMD+forex work; must be cited in Section 2.2
Sentiment + VMD paper (Expert Systems with Applications, 2025)	Sentiment + VMD for financial prediction; challenges Phase A+B joint novelty
Persistent Laplacian for stock markets (Entropy MDPI, 2023)	Direct prior art for TEPC topology block
ICML 2023 persistent Laplacian paper	Foundational ML method paper; must be cited
Chaos/forex papers (Springer ICSI 2014, Open Economies Review 2022)	Prior chaos+forex work
Anchoring bias in LLMs papers (JCES 2025, JBEF 2024)	Ground debiasing claims in Section 3.8
Evaluating LLMs in Finance (arXiv 2602.14233, 2026)	Directly relevant to Final-Phase bias-control motivation

Revised novelty framing. Replace "first ever" language with:

- "First to combine debiased multi-persona LLM prompting with GDELT thematic sentiment and GARCH-family volatility modeling for USD/INR prediction."
- "First to engineer coupled Lorenz oscillators across a time-varying financial graph as predictive features for a walk-forward forex classification task."
- "First to demonstrate, under a strict two-day embargo and identical input conditions, that a compressed memory snapshot outperforms frontier LLMs on USD/INR price MAE."

These are narrower but fully defensible.

R12. Address the Condorcet Independence Issue

Problem. The theoretical grounding (Section 3.8) claims the nine personas approximate independent agents, but they share a single model and agree 60–80% of the time.

Required action.

- Add a paragraph explicitly acknowledging that prompt-level differentiation does not produce statistical independence.
 - Reframe the Condorcet argument as an approximation: "To the extent that different prompts elicit attention to different input features, the personas achieve partial effective independence. The 60–80% agreement rate corresponds to an effective sample size of approximately 2–3 independent agents rather than nine — still above the majority-vote threshold, but the diversity benefit is more modest than the architecture implies."
 - Cite the Wisdom of Silicon Crowd paper (arXiv 2402.19379) for empirical evidence on LLM crowd accuracy.
 - Add an experiment measuring per-persona directional accuracy individually (not just as an ensemble) to test whether any persona has $p > 0.5$ as required by Condorcet.
-

R13. Correct the "Persistent Laplacian" Framing

Problem. The TEPC block does not implement a true persistent Laplacian.

Required action.

- Rename the block "Multi-Threshold Spectral Graph Filtration" or "Graph-Laplacian Filtration" throughout the paper.
 - In Section 3.9, replace "Persistent-Laplacian-style filtration" with: "We implement a graph-Laplacian filtration over edge-strength quantiles, which approximates the persistence intuition — tracking how network features survive across scales — without the full machinery of simplicial complexes and boundary operators required by true persistent Laplacians."
 - Cite the ICML 2023 persistent Laplacian paper and the Entropy 2023 financial persistent Laplacian paper, and explicitly position TEPC relative to them.
 - Note in the future work section that replacing this with a full persistent-Laplacian stack (as already mentioned) is a natural extension.
-

Tier 4: Writing and Presentation

R14. Rewrite the Abstract

Replace the current component-listing abstract with a four-part narrative abstract (as outlined in the writing review): problem statement, high-level approach, top three empirical findings, central lesson. Target 150–180 words.

R15. Add a System-Level Overview in Section 3.1

Add one paragraph and one figure that shows all three frameworks (Phase A–D, TEPC, Final-Phase) as a progression with increasing bias control stringency. This figure should replace or supplement Figure 1.

R16. Move Hyperparameter Details to an Appendix

All specific parameter values (learning rates, dropout, GARCH order notation, Lorenz constants) should move to "Appendix D: Implementation Details and Hyperparameters." Section 4.2 should retain only model family descriptions.

R17. Define the STABLE Threshold

State explicitly in Section 3.4 the return magnitude threshold used to define UP vs. STABLE vs. DOWN. This number affects every directional accuracy figure in the paper.

R18. Add a Cross-Correlation Figure for Lead-Lag

As described in R8, add a plot of cross-correlation vs. lag for political sentiment vs. IMF3 movements. This is the paper's most accessible empirical finding and currently has no figure.

R19. Fix Tables and Figures

- Make Figures 6 and 7 publication-quality vector graphics with readable axis labels.
 - Add a Final-Phase actual vs. predicted time-series figure for the MEMORY-ONLY arm.
 - Fix Table 21 feature importances to sum to 100%, or label it "Top 10 features (87.4% of total importance)."
 - Add a Glossary/Acronym appendix covering all abbreviations (TEPC, VMD, IMF, GDELT, MACRO-BASELINE, TOPOLOGY-ONLY, CHAOS-ONLY, MEMORY-ONLY, etc.).
-

R20. Consolidate Result Tables

Merge Tables 14 and 15 (two TEPC ablations) into a single table with a Window column.
Merge Tables 17 and 18 (100-day and 30-day Final-Phase) into a single table. This reduces the number of tables in Sections 5.9–5.10 from six to four and substantially reduces cognitive load.

Summary Roadmap

Tier	Issues	Type	Effort
1	R1–R5	Data/validity	Low–Medium
2	R6–R10	Experiments/analysis	High
3	R11–R13	Related work/framing	Medium
4	R14–R20	Writing/presentation	Low–Medium

The Tier 1 and Tier 2 fixes are non-negotiable for resubmission. Tier 3 changes the paper's framing but not its conclusions. Tier 4 improves the paper's chance of acceptance at a competitive venue like EMNLP where presentation quality is heavily weighted. A paper that addresses all four tiers and shows clean results under corrected walk-forward VMD and proper significance testing would be a credible EMNLP submission.